Lesson 4: Project: Automate One Recurring Report

Module 1 | 60 minutes | End-to-end report | Adults (18+) · ages 18–65

Objective: Automate one recurring report from data to delivery by joining a self-updating sheet, a prompt template and an emailed draft, with human checkpoints at every risky step and an honest time-saved log.

Success criteria

- I can map a report pipeline with three human checkpoints.
- I can join a self-updating sheet, a prompt template and a draft email to myself.
- I can catch a wrong number or invented reason at a checkpoint.
- I can keep a time-saved log and work out the payback honestly.

Key vocabulary

- **pipeline**: All the steps of a task joined in order, from data in to message out.
- **human checkpoint**: A planned stop where a person checks before the work moves on.
- **draft-to-me rule**: Automations send drafts only to you; you send the final version.
- **time-saved log**: A weekly record of minutes taken (including checks) and problems caught.
- **payback**: How many weeks of time saved it takes to cover the setup time.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Printed project brief
- Project files: weekly_report.js, l04_report_automation_kit.md, l04_teacher_demo_and_marking.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	“Assembly line” with and without inspectors (see Unplugged).
4 min	**Watch + poll**	Play the project video. Poll: “Which report eats most of your week?” (type it in the chat).
6 min	**Live demo**	Pipeline drawing, checkpoints A and B, the Copilot comments catch (12% vs 8.7%, invented reason), the draft email, a log row.
5 min	**Prompt Lab**	Prompt Lab: the report comments template. Test it in a chatbot with the practice numbers and check the percentage.

5 min	**Team games**	Quick competition: “Script or checkpoint?” sort and the pipeline order game in two teams; “Payback time” as a speed question.
20 min	**Project build (pairs)**	Build steps 1–6. Halfway check: “Show me checkpoint A and B passing.” Partners swap drafts at checkpoint C and hunt for one wrong number.
7 min	**Show-and-tell**	Three learners show their pipeline map, the mistake their checkpoint caught, and their payback weeks.
5 min	**Probe and submit**	New situation: the stale week. Submit the module project with the reflection and time-saved log.
3 min	**Wrap-up**	Exit ticket and preview of Module 2: AI for your role.

Unplugged warm-up

Assembly line (5 min): four volunteers form a line to make a “report” on paper: one writes numbers, one adds them, one writes a sentence, one “sends” it. Secretly tell the adder to make one mistake. Run it once with no inspectors (the mistake goes out), then again with an inspector after each person. Ask: “Where did the inspector catch it? Where should checkpoints go in your report?”

Teaching the idea

This is the Module 1 project. Learners join lesson 1 (templates), lesson 2 (self-updating dashboard) and lesson 3 (a script that emails you) into one recurring report, end to end. The key design idea is human checkpoints: after cleaning (A), after the numbers (B) and before anything leaves the inbox (C). The automation emails drafts to the learner; the learner forwards. The AI only sees dashboard numbers, never raw data. Measurement is honest: checking time and setup time count, and learners work out the payback in weeks. The caught mistake: the AI’s comments say “up 12%” (really 8.7%) and invent a reason.

Analogy: A report pipeline is like a factory line with quality inspectors: machines cut, stamp and pack, but an inspector checks at three points, and nothing leaves the gate without a signature.

Live demo: 1) Draw the pipeline on a whiteboard or slide: data in → clean (A) → numbers (B) → words → send (C) → log. 2) Open the Lesson 2 dashboard and show checkpoint A (all OK) and B (filter check ₹15,310). 3) Paste the comments prompt into Copilot (or any chatbot) with the numbers; check the percentage with =B3/B8-1 (8.7%) and look for any added reason. If the AI is accurate, show the video version. 4) Run weekly_report.js on a copy; open the DRAFT email and read the “check before forwarding” line. 5) Fill a time-saved log row: 75 before, 20 after, setup 150, payback about 3 weeks. 6) Show the rubric and the project requirements.

Questions to ask

Ask	Listen for
Where do the three checkpoints go?	After cleaning, after the numbers, and before anything is sent to others.

Why does the script email only you?	So a person checks the draft before it reaches anyone else.
Why give the AI dashboard numbers and not the raw data?	Less private data shared, and fewer chances for wrong totals; you can check every number.
How do you work out payback?	Setup minutes divided by minutes saved per week.
What goes in the time-saved log?	Date, minutes before/after including checks, problems caught.

Differentiation

Support
Offer the complete practice pipeline (Lesson 2 dashboard, the comments template and the checked script) so learners only run it, find the mistake at checkpoint C and fill the log.

Stretch
Feed the Orders tab from a Google Form so new orders arrive by themselves, add a "sales by item" table to the email, or set a weekly trigger for Monday morning.

Common misconceptions

Misconception	How to fix it
End to end means no human at all.	It means every step is joined up; people still check at A, B and C.
Time saved = old time − AI time.	Count checking and setup time too; then work out payback.
The AI needs all my data to write a good comment.	It needs only the summary numbers, and you can check them all.
Once it works, it will always work.	Dates, tab names and data change; checkpoints catch the drift.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Pipeline design and checkpoints	Steps unclear; no checkpoints	Steps listed; checkpoints vague or missing one	Clear pipeline with checkpoints A, B and C and who does each	Clear, efficient pipeline; checkpoints placed where mistakes would spread, with a routine card a colleague could follow

Automation quality	Numbers typed by hand each time	Some formulas, but the dashboard does not update from one date cell	Dashboard updates from one cell; template and draft email work	Everything updates and drafts reliably; script checked with all 7 points (or a clean template alternative)
Checking and accuracy	AI numbers or comments used unchecked	Some numbers checked	Every number in the draft checked; one caught mistake explained	Checked, caught mistake explained, and a fix added so it cannot happen again
Measurement and responsible use	No times; private data used	Times without checks or setup	Honest before/after with checks, setup and payback; no private data in AI	Honest log over two runs, payback clear, privacy and draft-to-me rule explained for a manager

Exit ticket
Name your three checkpoints, and say which one you think will catch the most problems in your report.

Homework
Finish the module project and run your pipeline for real next week. Add one more row to your time-saved log.

Answer key — Kitabash activities

1. Prompt Lab: the report comments template (perform · ai lab)

Checklist: Say who it is for · Use only the numbers given · Include this week and last week · Ask for the change, correctly · Forbid added reasons or numbers · Set the length · Ask it to flag anything unclear

Strong example:

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You write my weekly sales note for my manager. Use only these numbers: week of [Week];
delivered sales [Sales] (last week [Last week]); [Orders] delivered orders; [Cancelled]
cancelled; [Pending] pending; credit to collect [Credit]. Write 3 short lines. Say
whether sales went up or down and give the percentage change from last week. Do not add
reasons, guesses or numbers that are not given. If anything looks unclear or missing,
flag it at the end instead of guessing.
```

2. In your report pipeline: let the sheet or script do it, or a human checkpoint? (perform · categorize)

Let the sheet or script do it: Adding up delivered sales with SUMIFS, Recalculating when the week date changes, Building the draft email from dashboard cells, Colouring sales red when below target
Human checkpoint: Confirming every row says OK before using the numbers, Reading the AI's comments against the dashboard, Forwarding the final report to the manager, Deciding what to do about a big drop in sales

3. The report pipeline (perform · ordering)

Order: Data arrives in the Orders tab → Checkpoint A: every row says OK → Dashboard updates from the week cell → Checkpoint B: one number verified with a filter → AI drafts comments from the dashboard numbers → Script emails the draft to you → Checkpoint C: read, fix and forward → Log the time and any problem caught

4. Payback time (perform · multiple choice)

Answer: 4 weeks

5. Module project: automate one recurring report (probe · project)

6. New situation: the stale week (probe · multiple choice)

Answer: Checkpoint C catches it; the lasting fix is a B1 formula for last week's Monday, =TODAY()-WEEKDAY(TODAY(),3)-7, or a first step on the routine card: "update B1"

Answer key — worksheet

1. **Key words:** pipeline = All the steps of a task joined in order, from data in to message out. · human checkpoint = A planned stop where a person checks before the work moves on. · draft-to-me rule = Automations send drafts only to you; you send the final version. · time-saved log = A weekly record of minutes taken (including checks) and problems caught.
2. **Prompt Lab: the report comments template:** Include: Say who it is for; Use only the numbers given; Include this week and last week; Ask for the change, correctly; Forbid added reasons or numbers; Set the length; Ask it to flag anything unclear. Example: You write my weekly sales note for my manager. Use only these numbers: week of [Week]; delivered sales [Sales] (last week [Last week]); [Orders] delivered orders; [Cancelled] cancelled; [Pending] pending; credit to collect [Credit]. Write 3 short lines. Say whether sales went up or down and give the percentage change from last week. Do not add reasons, guesses or numbers that are not given. If anything looks unclear or missing, flag it at the end instead of guessing.
3. **In your report pipeline: let the sheet or script do it, or a human checkpoint?:** Let the sheet or script do it: Adding up delivered sales with SUMIFS, Recalculating when the week date changes, Building the draft email from dashboard cells, Colouring sales red when below target · Human checkpoint: Confirming every row says OK before using the numbers, Reading the AI's comments against the dashboard, Forwarding the final report to the manager, Deciding what to do about a big drop in sales
4. **The report pipeline:** 1 Data arrives in the Orders tab · 2 Checkpoint A: every row says OK · 3 Dashboard updates from the week cell · 4 Checkpoint B: one number verified with a filter · 5 AI drafts comments from the dashboard numbers · 6 Script emails the draft to you · 7 Checkpoint C: read, fix and forward · 8 Log the time and any problem caught
5. **Payback time:** B. 4 weeks — You save 70 − 25 = 45 minutes a week. 180 ÷ 45 = 4 weeks. After that, every week is pure time saved.
6. **My plan:** A clear plan that covers every requirement.
7. **Checklist:** All requirements met.
8. **Two stars and a wish:** Kind, specific feedback and an improvement.

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